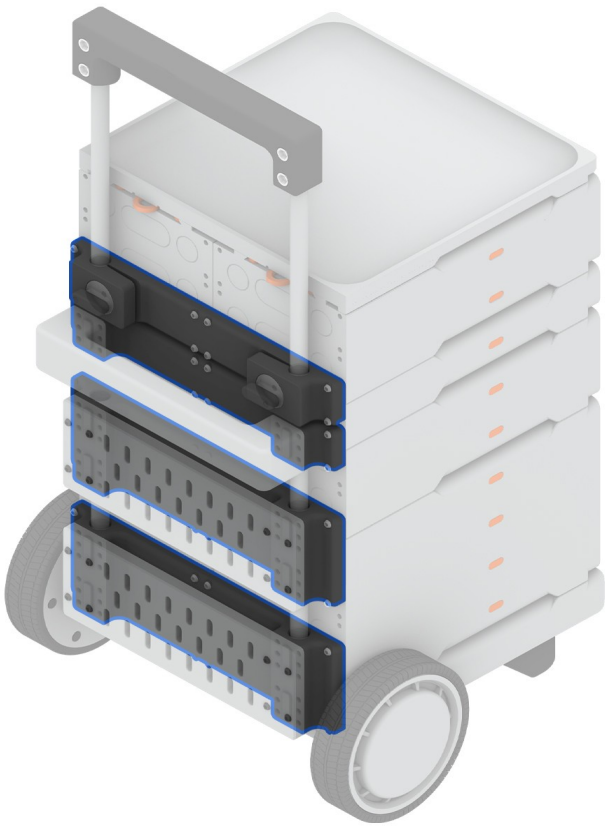
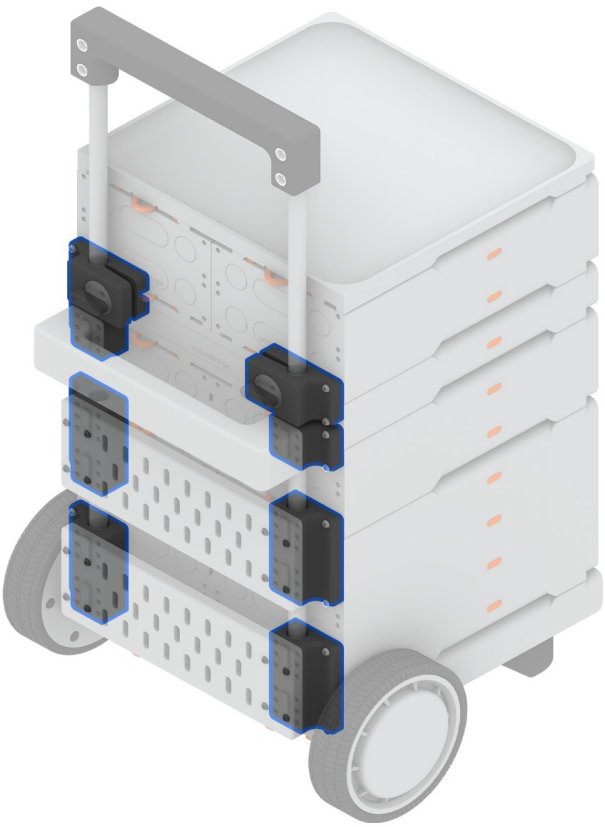
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Modifications Assembly 1/15

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this is a tower made from L and S modules of different heights,
we can use both wide L-width backbone parts,or universal parts

·· 1

·····

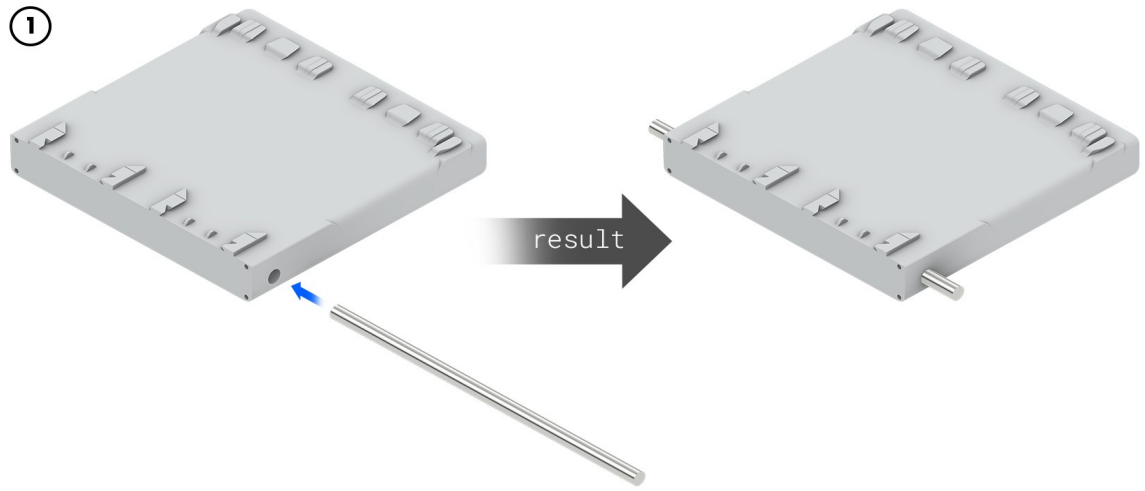
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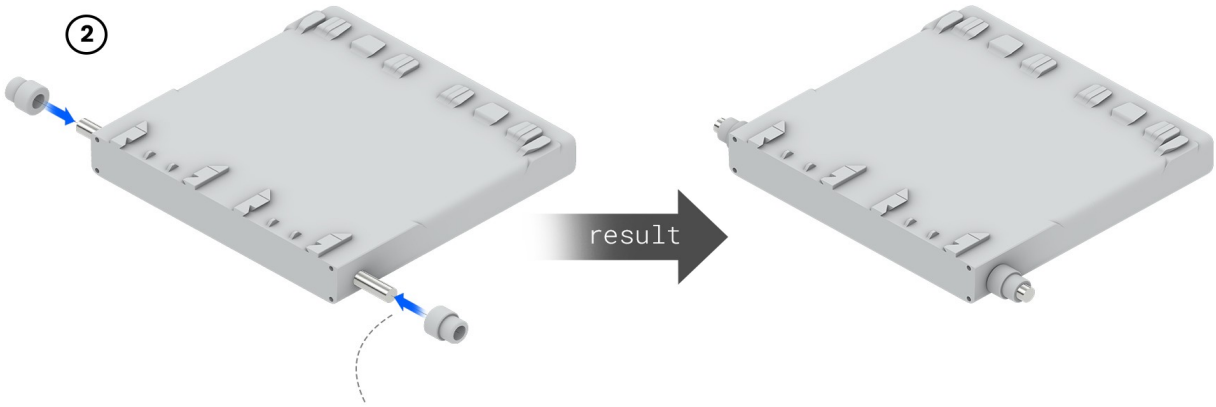
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Modifications Assembly 2/15

#ussadesign





*check the **next page** to see how to **insert** this **correctly***

1. cut a rod to 504MM for XL, or 414MM=L, 338MM=M and 248=S, and push it into a base
2. use a special jig to push the bushings from both sides on a rod *(see next page)*

.. 2 ·····

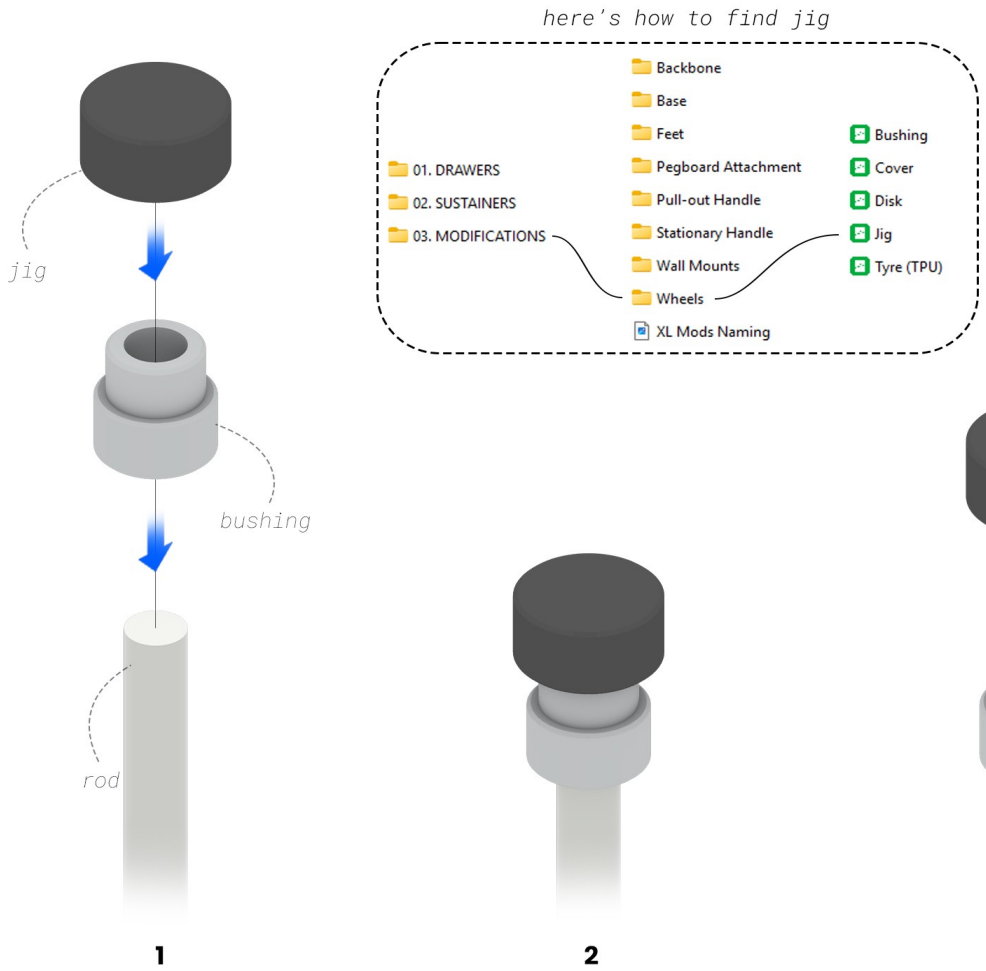
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Modifications Assembly 3/15

#ussadesign



1. push bushing in first, then a jig, to have 10MM of a rod sticking out of the bushing
2. it should look like this, after the first step
3. take out the jig, after that

·· 3

M8 ·····

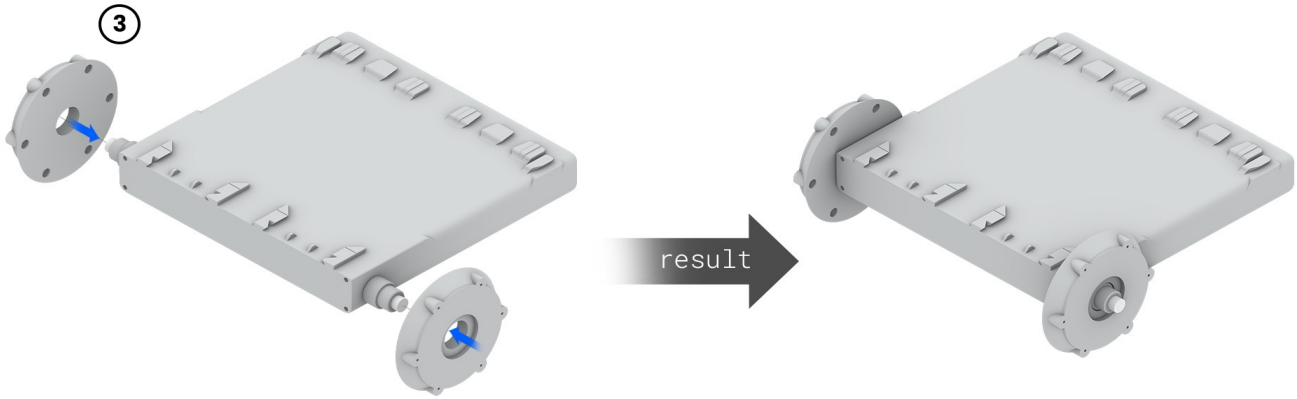
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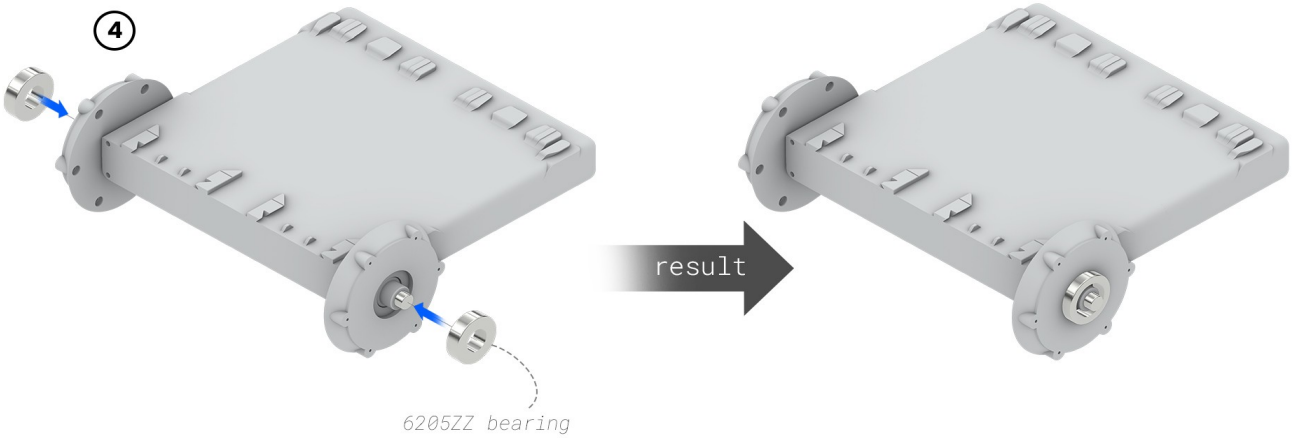
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Modifications Assembly 4/15

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3. place disks from both sides of the rod, it should get on a bushing
4. push bearings (6205ZZ, 25d-52D-15MM thickness) on printed bushings

·· 4 ·····

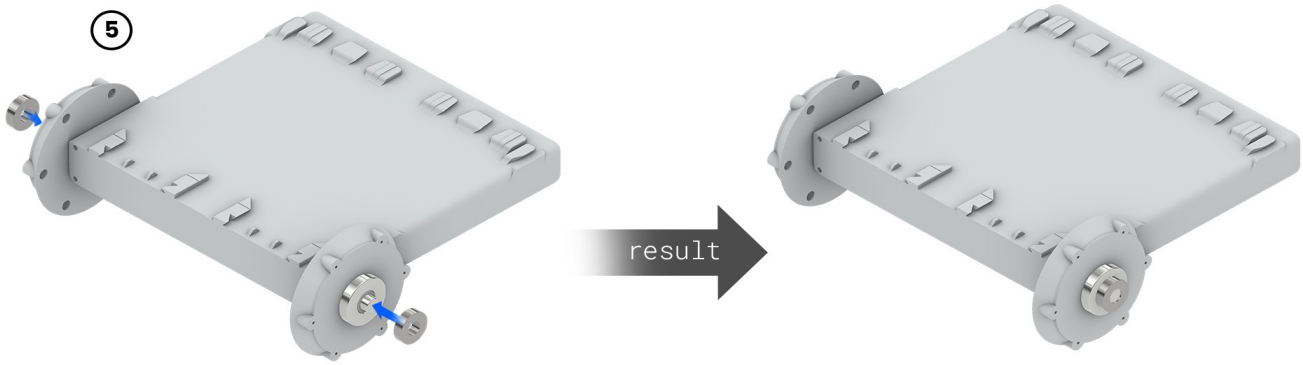
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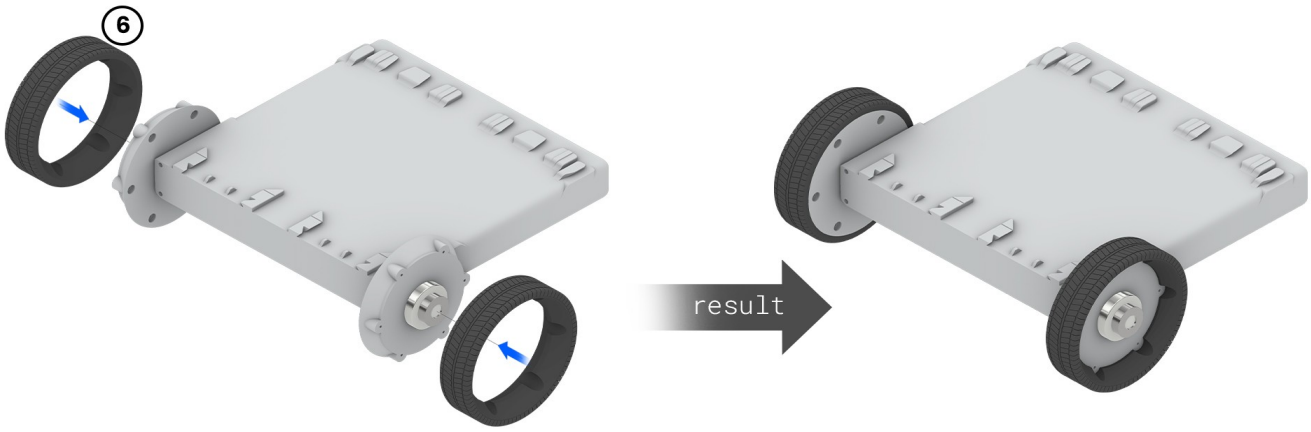
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Modifications Assembly 5/15

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5. after you get bearings on, we need to stop them with SCSAWs (16d, 30D, 10MM thick)
6. we recommend using TPU for the tires, after printing, slide them on disks

·· 5

M16 ·····

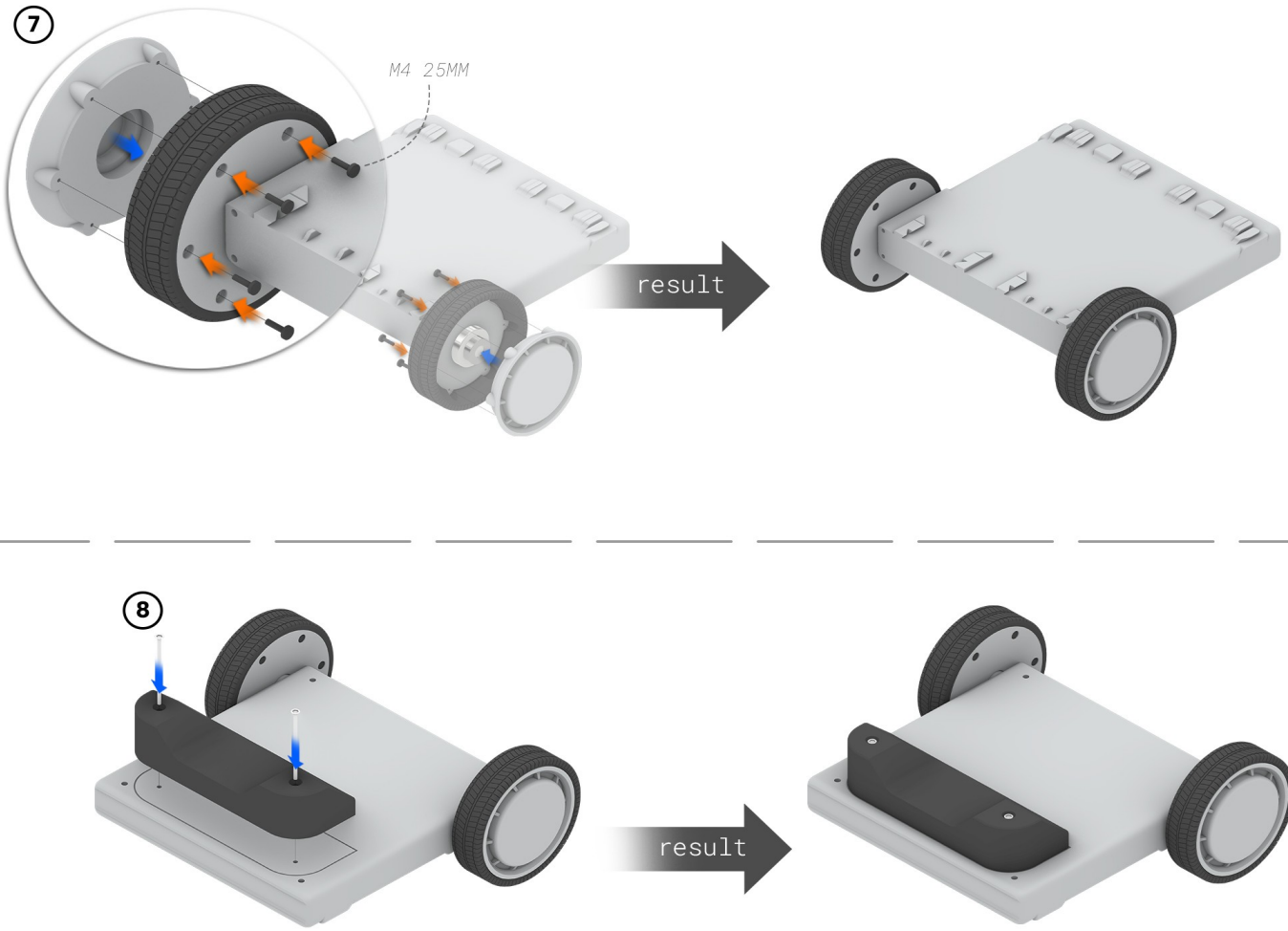
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Modifications Assembly 6/15

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7. use a "cover" part to close it up with 6 M4 25MM bolts per wheel
8. flip it upside down and attach the necessary size foot with M6 90MM

·· 6

····· TPU ··

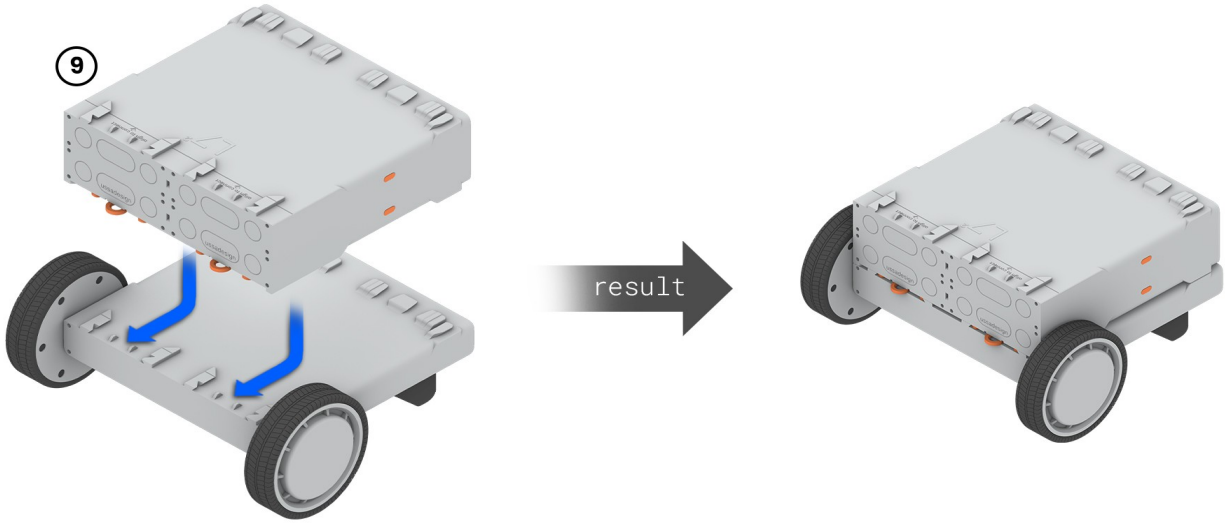
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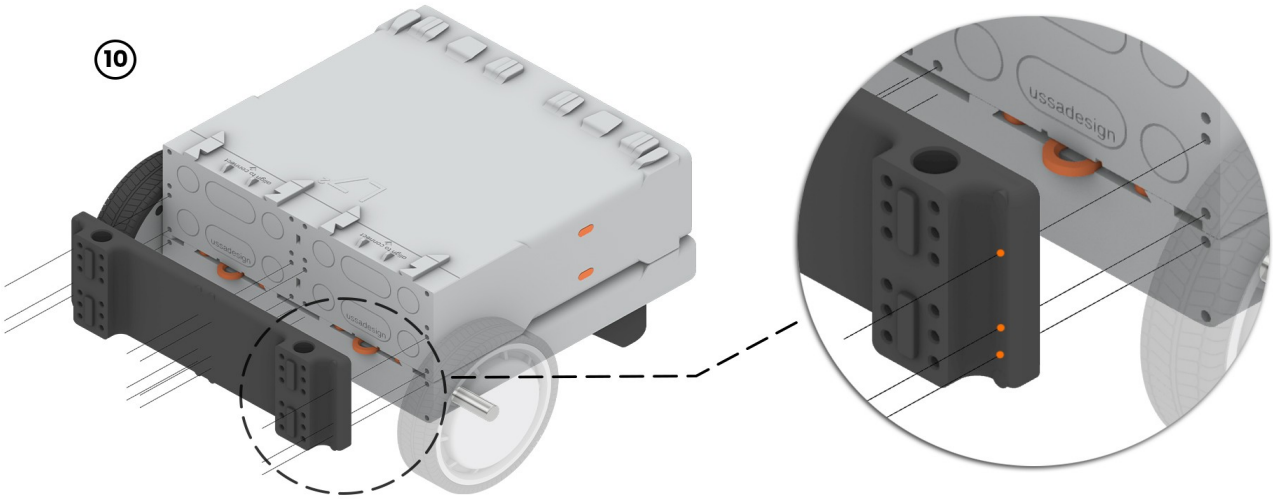
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Modifications Assembly 7/15

#ussadesign





9. WE HIGHLY RECOMMEND USING 2F MODULE AFTER BASE

10. prepare a 2F backbone for drilling holes, they come sealed. Mark the three spots

·· 7

10mm ····· (Jig)

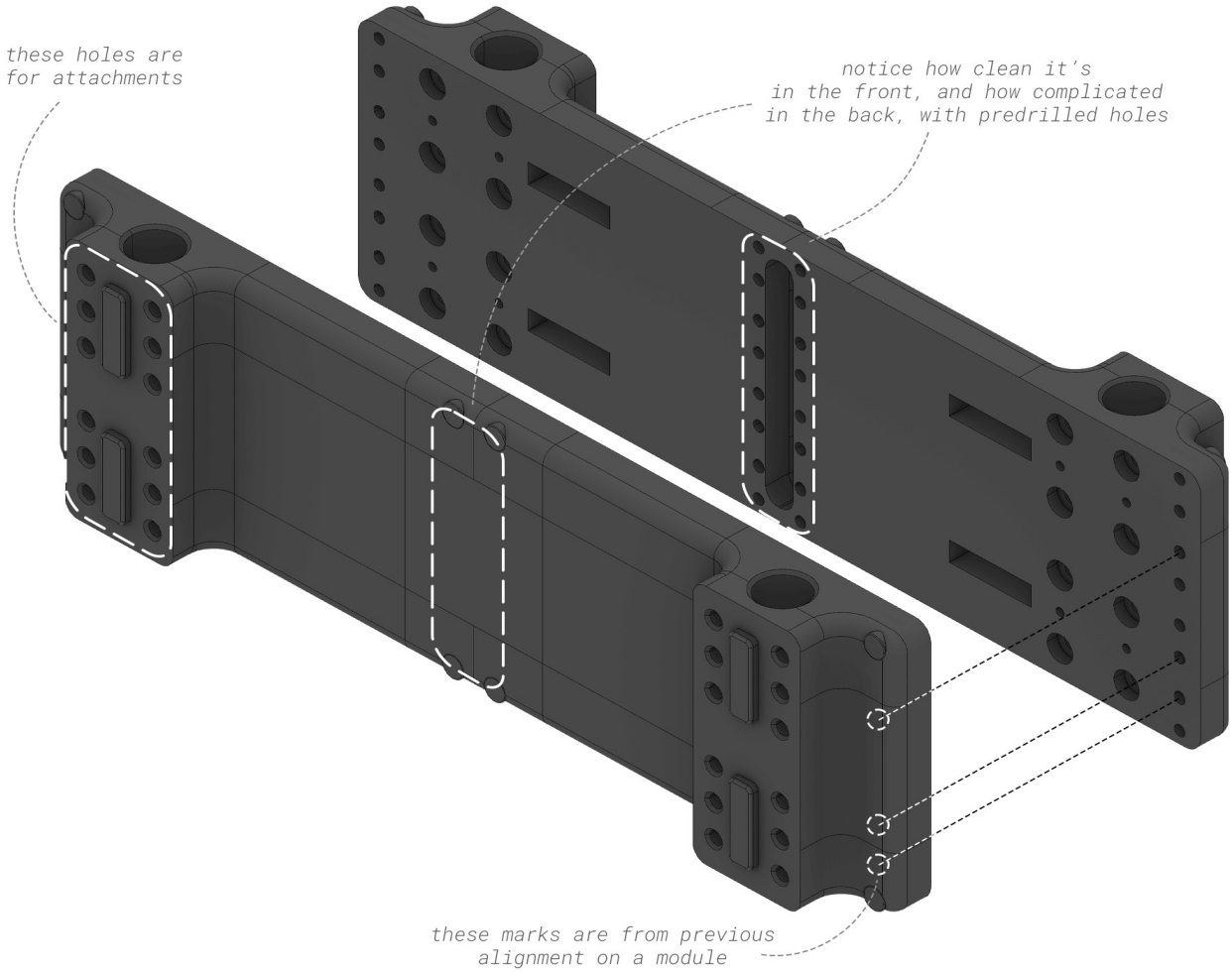
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Modifications Assembly 8/15

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after you mark necessary spots, you can go ahead
and drill them through with 4mm bit

·· 8 ·····

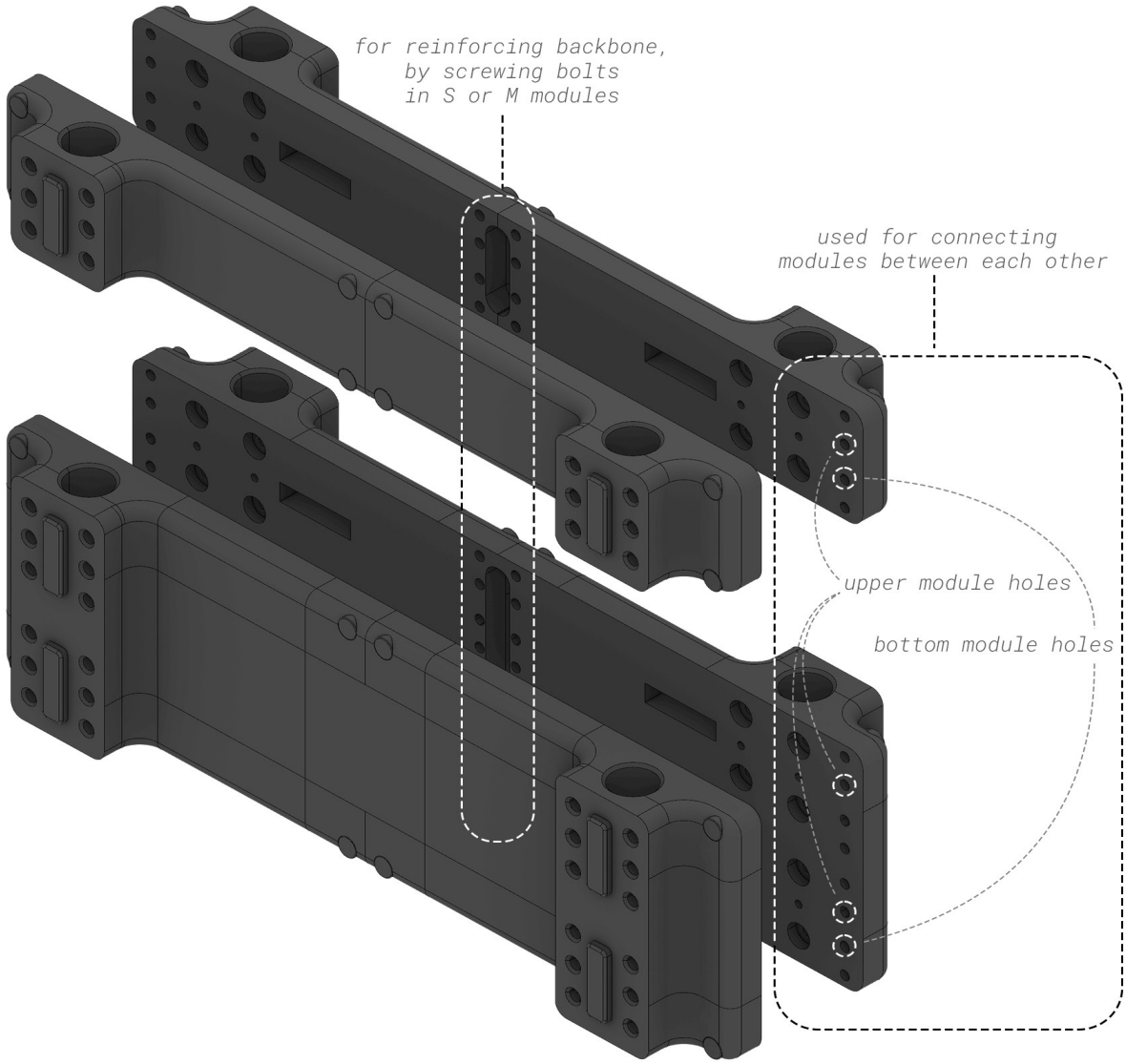
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Modifications Assembly 9/15

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·· 9

····· (Backbone) ··

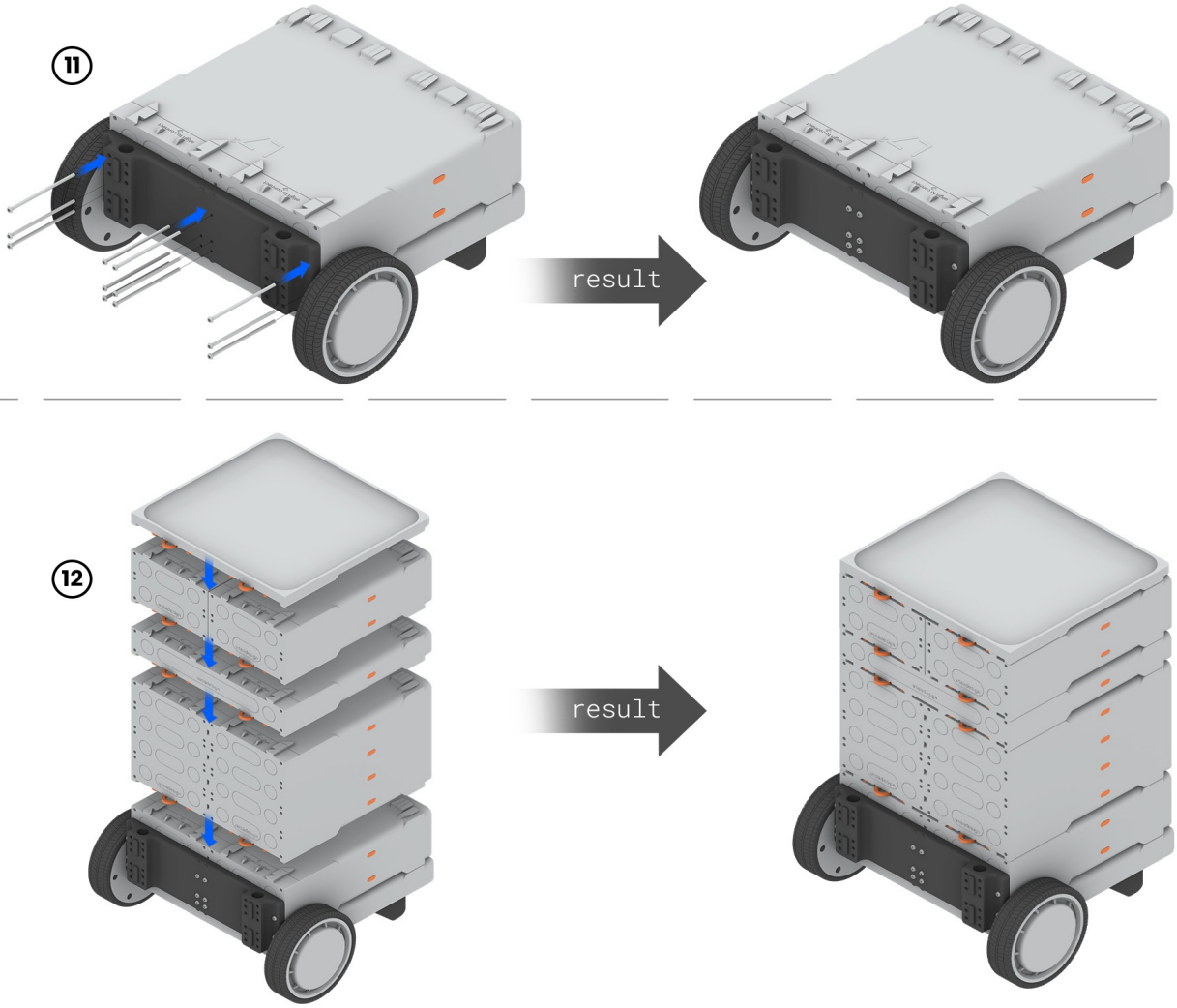
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Modifications Assembly 10/15

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11. now you can attach the backbone both to base and module
12. WE HIGHLY RECOMMEND ENDING WITH 2F MODULE BEFORE ROOF

·· 10 ·····

·· ·····

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Modifications Assembly 11/15

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13. connect the remaining modules with each other
14. you can use 1F backbone for connecting modules as well

·· 11

····· (M4 × 60 mm)

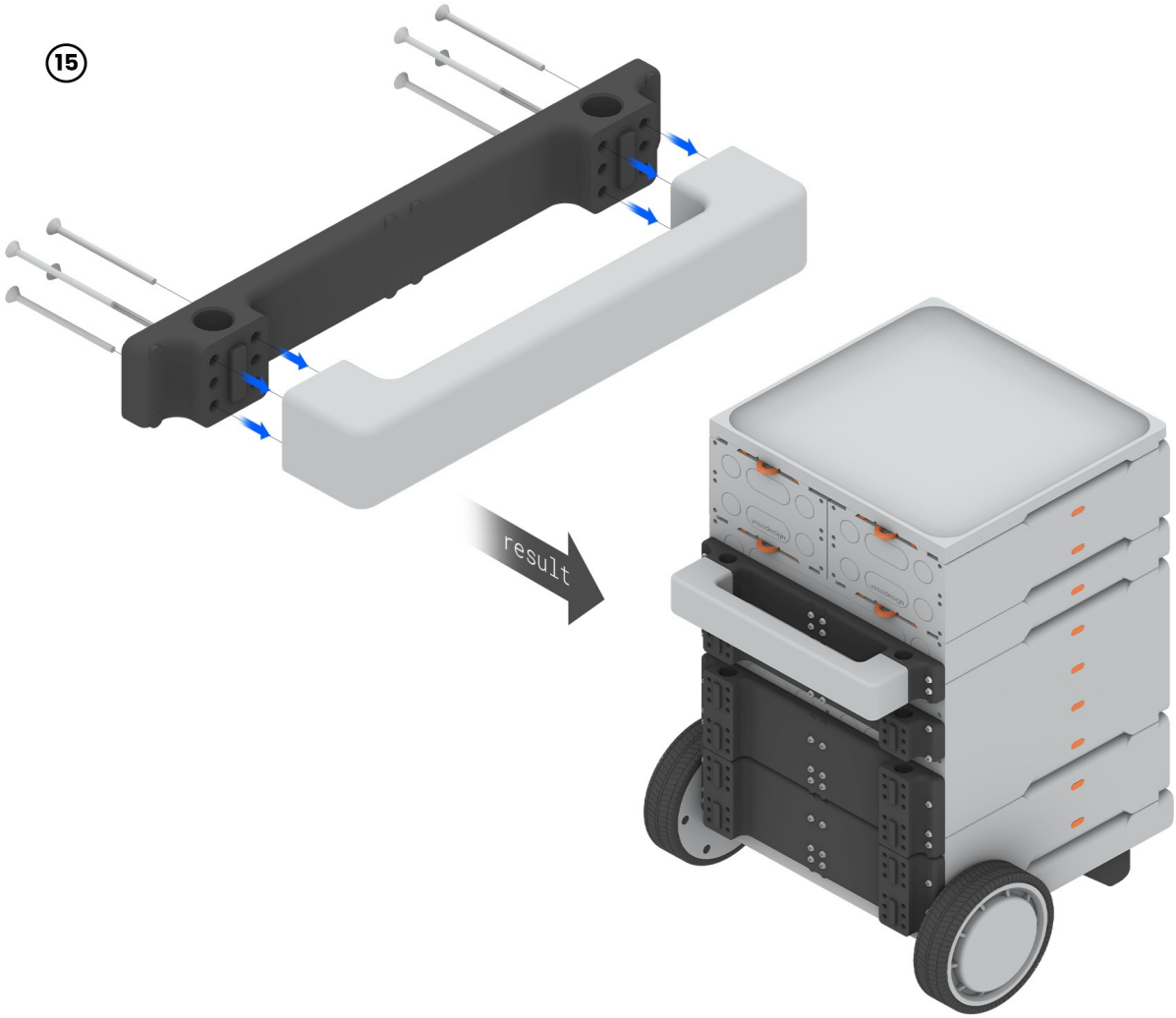
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Modifications Assembly 12/15

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15. if you want to use a stationary handle, you need to attach it from the other side of a backbone, before attaching backbone part to a module

·· 12

·····

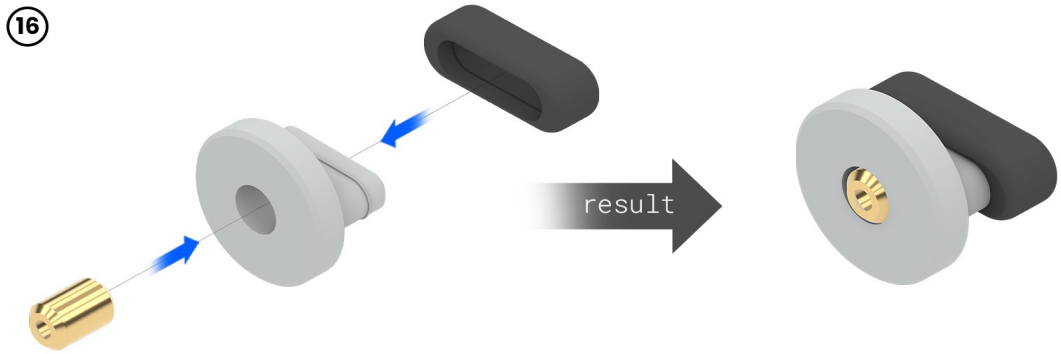
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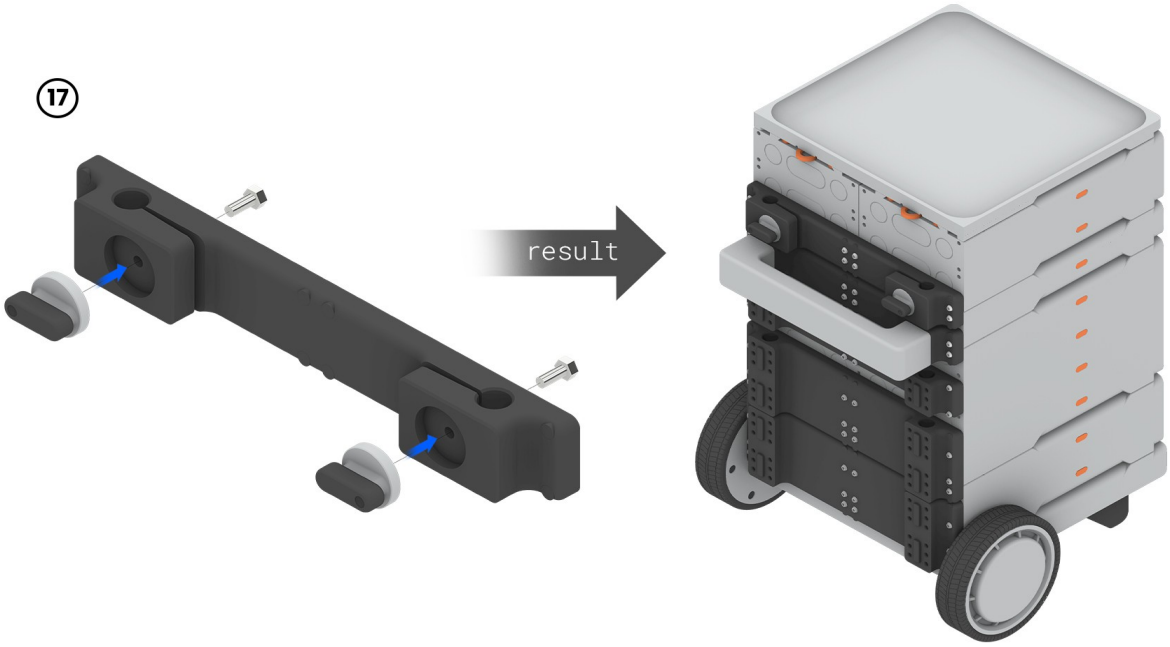
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Modifications Assembly 13/15

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16. heat up heat insert, and push it straight in a “knob”, then cap it
17. use Hex M4 35MM bolts from another side of a backbone to tighten the lock

·· 13

···· M6 ·····

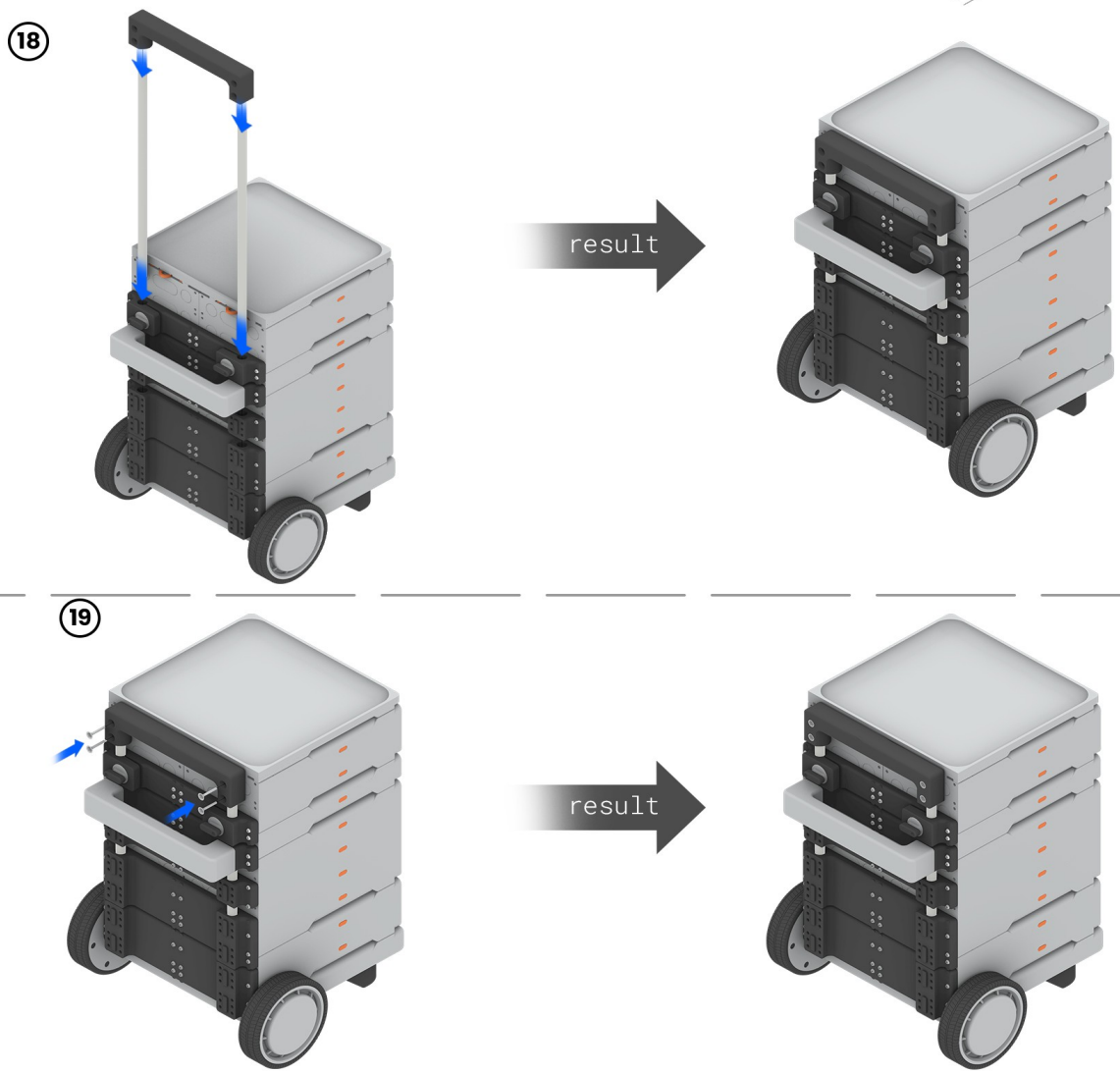
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Modifications Assembly 14/15

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18. cut 2 M16 rods to the height of your tower, and slide them along with a handle
19. drill M6 holes and tap the rods to attach the pull-out handle

·· 14

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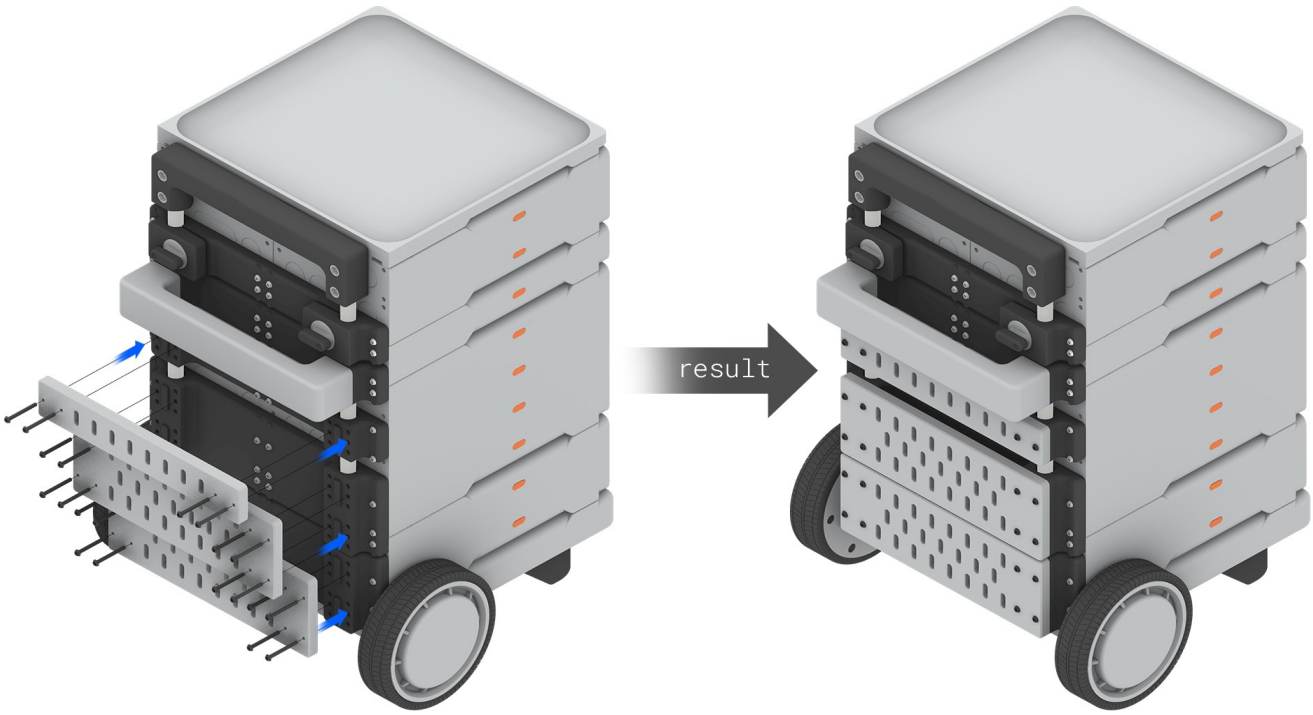
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Modifications Assembly 15/15

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20



20. attach IKEA or USSA pegboards on backbones, USSA pegboard link: https://www.ussadesign.com/products/multifunctional-bench

·· 15

····· (Pegboard) ····

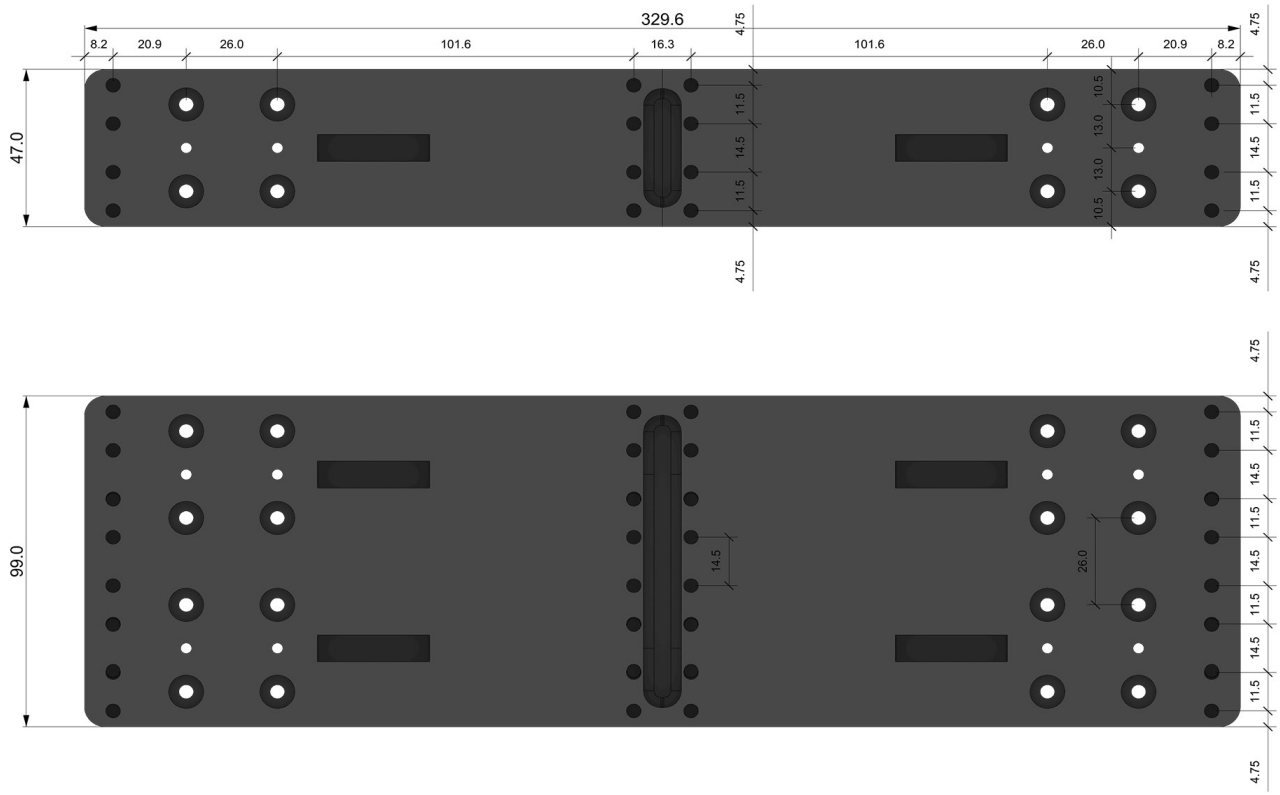
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1F&2F L Backbone Dimensions

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·· 16

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Casters Mode

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flip the base, insert 4 casters with M8 bolts and flip it back

·· 17

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